Last week:

- Spanning trees.
- Minimum spanning trees in weighted graphs:
  - Kruskal's algorithm
  - Prim's algorithm

This week:

- Counting spanning trees (no proofs).
- Measuring connectivity:
  - Cut-vertices
  - Blocks

Recall: • A spanning tree of a graph G is a subgraph of G which is spanning (uses all vertices) and is also a tree (connected, no cycles).

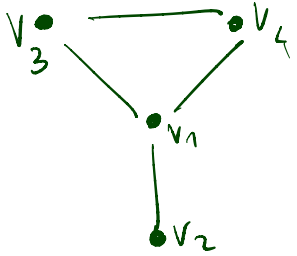
• If $V(G) = \{v_1, \dots, v_n\}$, the **incidence matrix** of G is $A = [a_{ij}]$ where $a_{ij} = \begin{cases} 1 & \text{if } v_iv_j \in E(G), \\ 0 & \text{otherwise.} \end{cases}$

Definition. • *The **Laplace matrix** of G is $Q = [q_{ij}]$ where* $q_{ij} = \begin{cases} \deg_G(v_i) & \text{if } i = j, \\ -1 & \text{if } v_iv_j \in E(G), \\ 0 & \text{otherwise.} \end{cases}$

• *We denote by $Q(1,1)$ the matrix obtained from Q by deleting the first row and column.*

Theorem. *(The Matrix Tree Theorem) If G is a connected graph with Laplace matrix Q, then the number of spanning trees of G equals $\det Q(1,1)$.*

Example. *Compute the number of spanning trees of the following graph.*



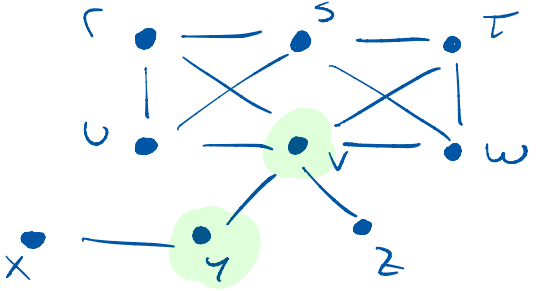
$$Q = \begin{bmatrix} 3 & -1 & -1 & -1 \\ -1 & 1 & 0 & 0 \\ -1 & 0 & 2 & -1 \\ -1 & 0 & -1 & 2 \end{bmatrix}, \quad Q(1,1) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

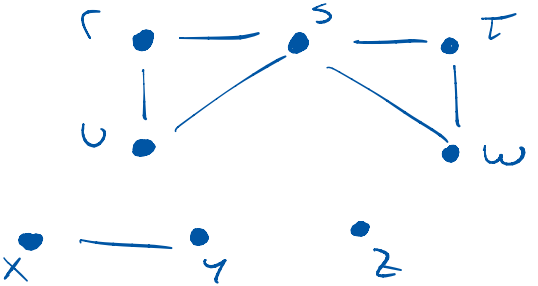
$$\det Q(1,1) = 4 - 1 = 3.$$

Recall: • An edge e in a graph G is a **bridge** if $G - e$ has more components than G.
• If v is a vertex in G, then $G - v$ deletes v and all incident edges from G.

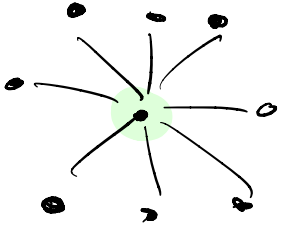
Definition. *A vertex v in a graph G is a **cut-vertex** if $G - v$ has more components than G.*

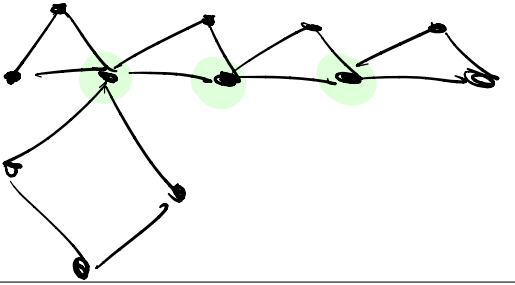
Example. *Identify all cut-vertices in the following graph.*





Example. *(1) Many bridges and a unique cut-vertex.* *(2) Many cut-vertices but no bridges.*





Theorem (5.1). *Let v be a vertex incident with a bridge in a connected graph G. Then v is a cut-vertex of G if and only if $\deg v \geqslant 2$.*



Proof: Let $e = uv$ bridge.

$\Rightarrow$) Assume v is a cut-vertex. If $\deg v = 1$ Then v is a leaf

So $G-v$ is connected, contradicts v cut-vertex.

$\Leftarrow$) Assume $\deg v \geq 2$. Let $w \neq u$ another neighbor for v.

WTS: $G-v$ is disconnected. Let's show u, w are in different components of $G-v$. Assume There is a u-w path P in $G-v$

Then P is a u-w path in G not using v, nor e.

Hence P is a u-w path in $G-e$, contradicts e is a bridge!

Exercise 1. *In a tree with at least 3 vertices, any non-leaf is a cut-vertex.*

Proof: In a Tree, all edges are bridges.

So all vertices are incident To some bridge.

Now use Theorem → Corollary 5.2

Exercise 2. *Let* G *be a connected graph on at least 3 vertices. If* G *contains an ~~edge~~ bridge then it contains a cut-vertex.*

Proof: The bridge looks like $e=uv$

AT least one of Them is a non-leaf (because $|V(G)| \geq 3$)

That one is a cut-vertex (Theorem)



Theorem (5.3). *Let v be a cut-vertex in a connected graph G, and let u, w be vertices in distinct components of $G - v$. Then v lies on every u–w path of G.*

Proof: We know: There are no u-w paths in $G-v$

Thus every u-w path in G uses v.



Theorem (5.5). *Let* G *connected with at least* 2 *vertices and fix* u *in* V(G). *If* v *is a vertex that is farthest from* u *in* G, *then* v *is not a cut-vertex of* G.

Proof: For a contradiction, assume v is a cut-vertex.

Let $w \in V(G-v)$ in a different component from u.

Then every u-w path in G uses v. Then

$d(u,w) = d(u,v) + d(v,w) > d(u,v)$ contradicts def^n of v.

Exercise 3 (Corollary 5.6). *If* G *is connected with at least* 2 *vertices, then* G *contains at least* 2 *vertices that are not cut-vertices.*

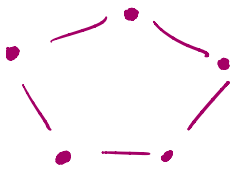
Proof:

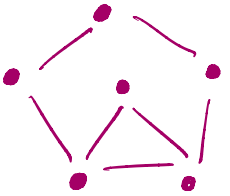
Let u,v such that $d(u,v) = \operatorname{diam} G$.

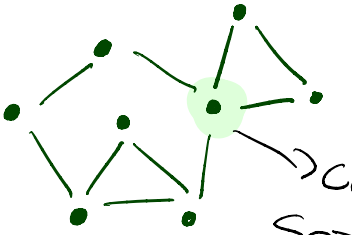
Then u is the farthest from v
v is the farthest from u
} u,v not cut-vertices.

Definition. *A connected graph with no cut-vertices is called* ***non-separable.***

Example: non-separable graphs:







Exercise: Assume G connected and $|V(G)| \geq 3$.

If G is non-separable $\Longrightarrow$ Every edge lies on a cycle.

Every pair of adjacent vertices lie on a cycle.

Proof: non-separable $\Rightarrow$ no cut-vertices $\Rightarrow$ no bridges

$\downarrow$

5.2 $\Rightarrow$ every edge lies on a cycle.

Definition. *A connected graph with no cut-vertices is called* ***non-separable****.*

Theorem (5.7)**.** *A graph* G *with* $|V(G)| \geqslant 3$ *is non-separable if and only if every two vertices lie on a common cycle.*

$\Leftarrow$) Assume every two vertices lie on a cycle.

WTS: G non-separable $\longrightarrow$ Connected ✓ (cycle $\Rightarrow$ 2 paths)
$\searrow$ no cut-vertices

Connected + no bridges $\Rightarrow$ no cut-vertices.

$\Rightarrow$) Assume G non-separable

WTS: every two vertices lie on a common cycle

(We know this is true for adjacent vertices)

Assume this is not true. Take u,v the closest vertices not in a common cycle. Let $k = d(u,v)$, Thus $k \geq 2$

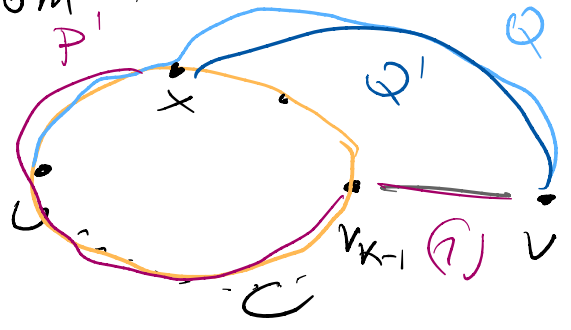
Take $P = (u = v_0, v_1, \dots, v_{k-1}, v_k = v)$ a u-v path.

Since $d(u, v_{k-1}) = k-1 < k$, there is a cycle C containing u, v_{k-1}

Since v_{k-1} is <u>not</u> a cut-vertex, there is a v-u path Q <u>not</u> using v_{k-1}.

Let x denote the first vertex in $Q \cap C$

Take Q' as the portion of Q going from v to x



Let P' be the v_{k-1}-x path on C using u.

We get a cycle C':

1) Go from v to v_{k-1}
2) From v_{k-1} to x along P'
3) From x to v along Q'.

cycle containing v and u!

□

Recall: Components in a graph G are equivalence classes under the relation on $V(G)$ defined by $u \mathcal{R} v$ iff there is a u–v path.

Definition. *Let G be a graph. We define a relation $\mathcal{B}$ on $E(G)$ by $e \mathcal{B} f$ iff $e = f$ or there is a cycle on G that uses e and f.*

Theorem (5.8)**.** *The relation $\mathcal{B}$ defined above is an equivalence relation.*